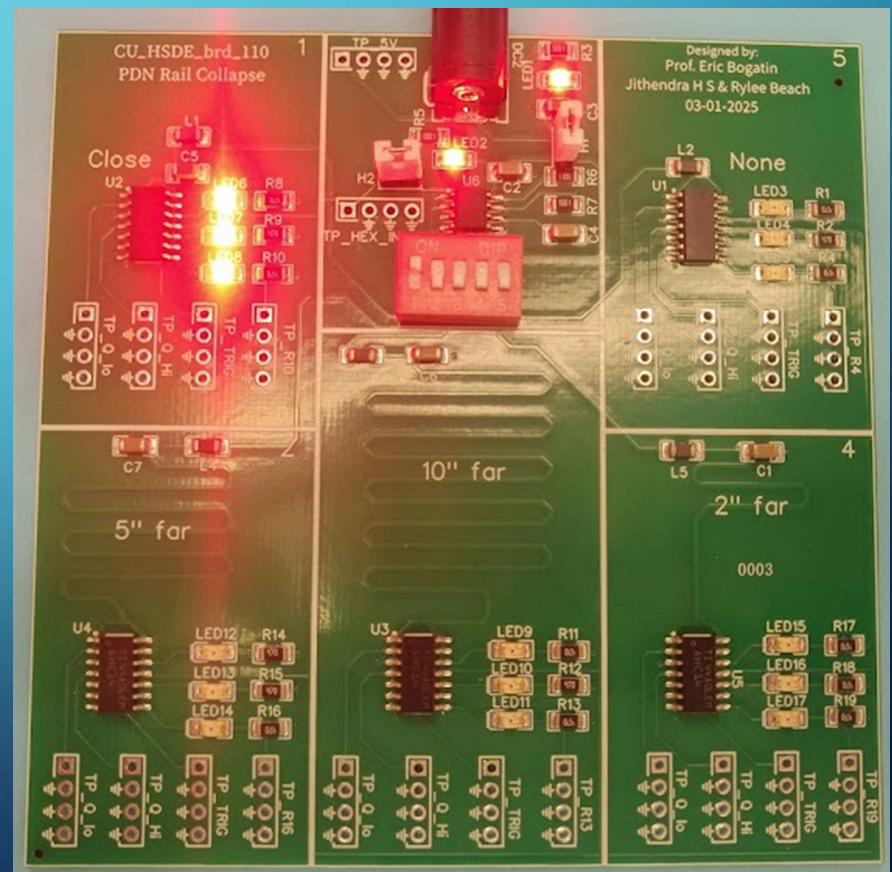
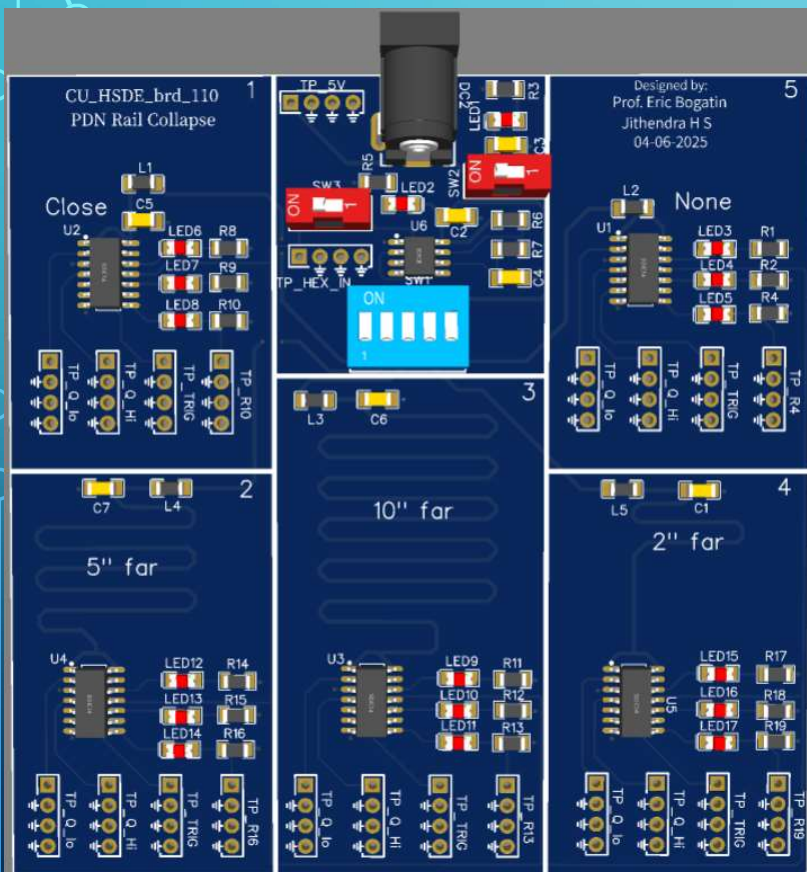


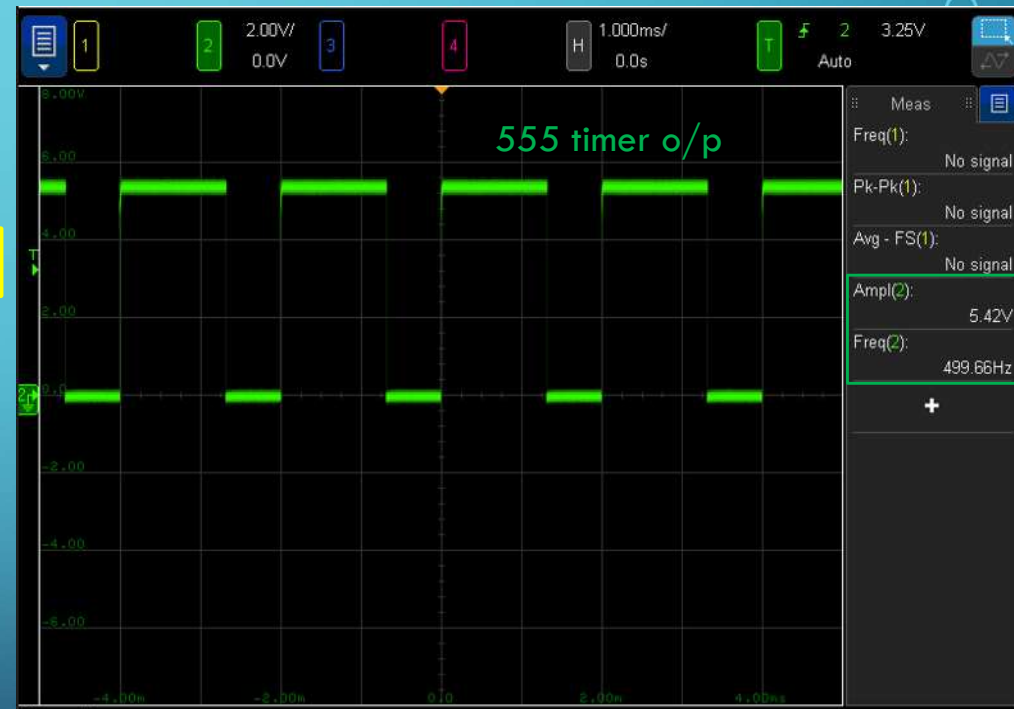
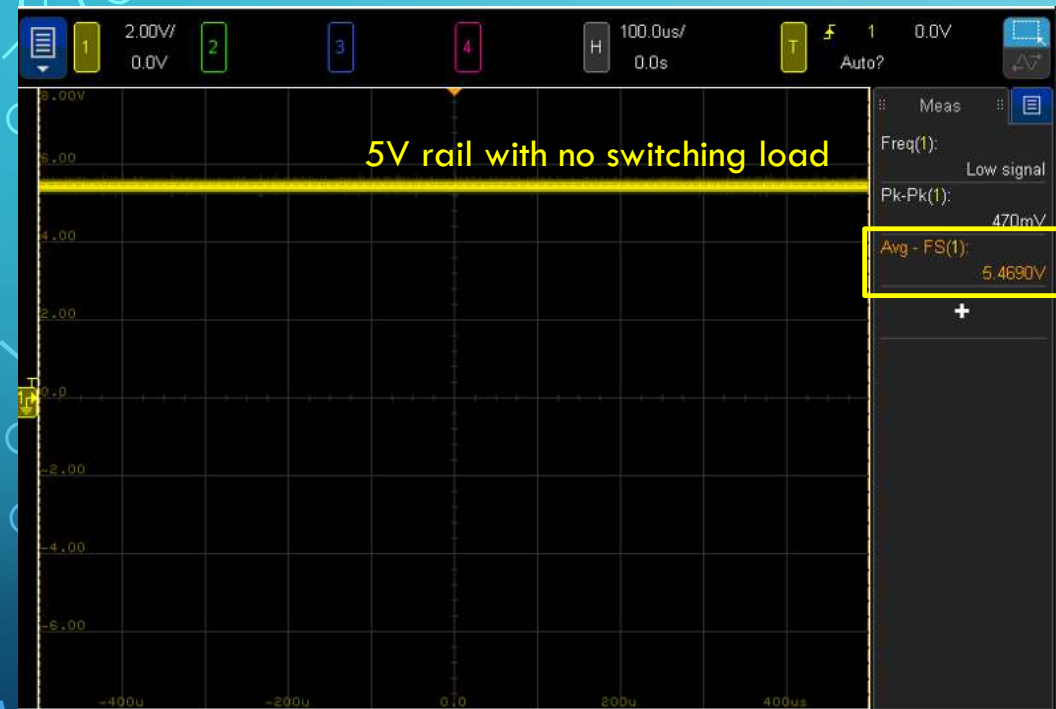
PRJ 110: PDN RAIL COLLAPSE



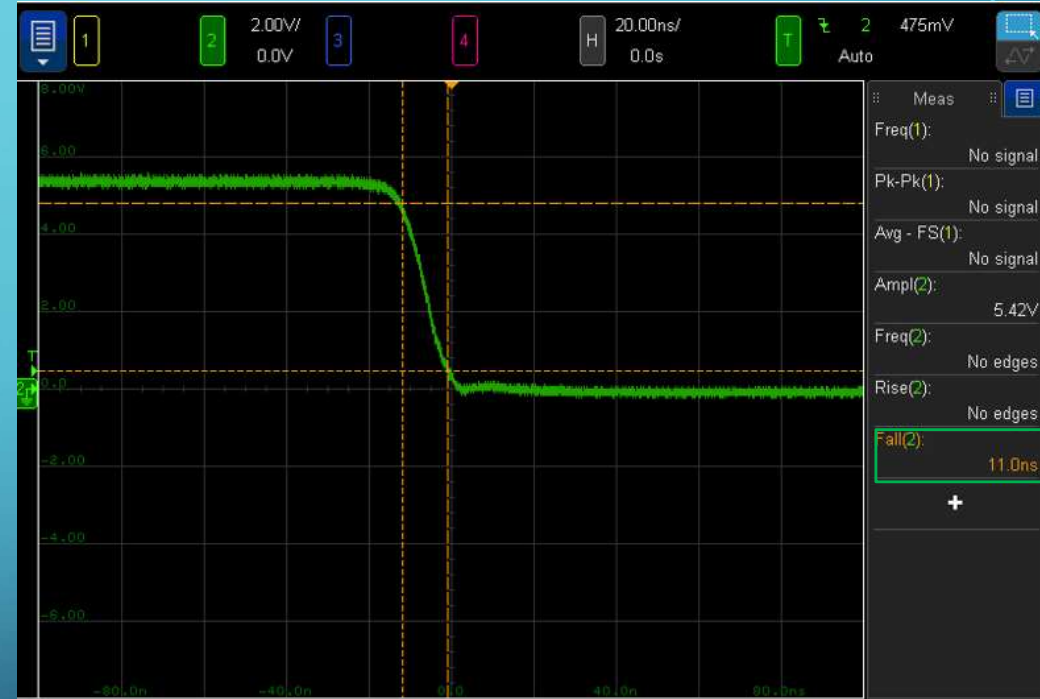
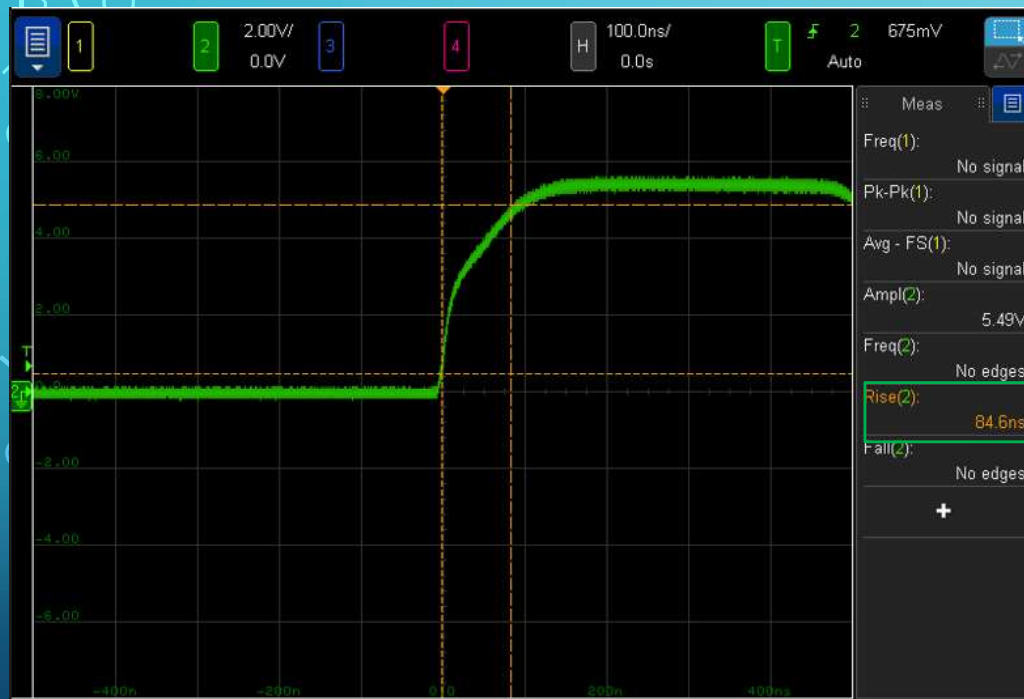
CHARACTERIZE

- 5V and 555 timer
- R_{th} of the inverter
- Current through LED
- Ferrite bead filter effect
- Q_{hi} and Q_{lo} noise comparison Table
- Rise time comparison

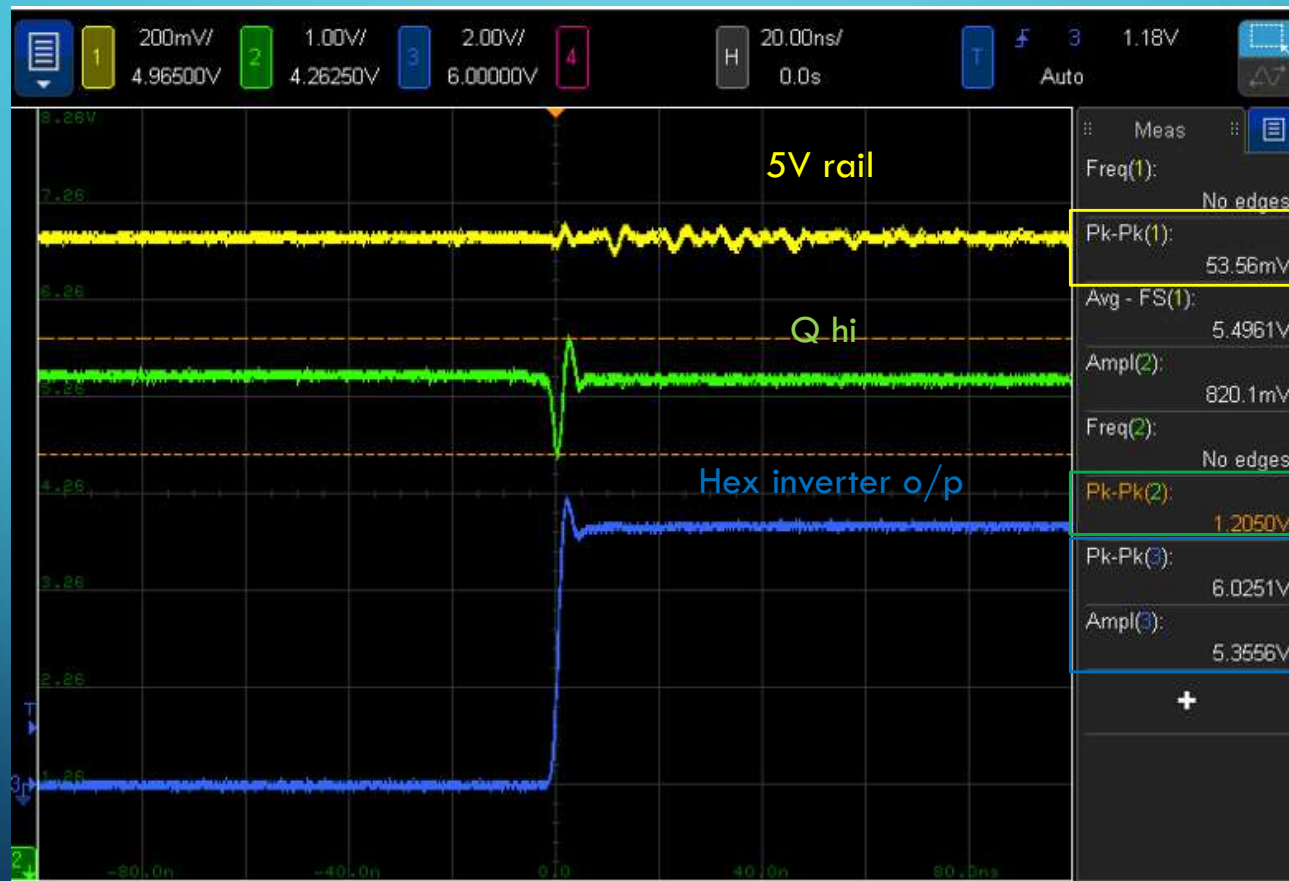
5V AND 555 TIMER



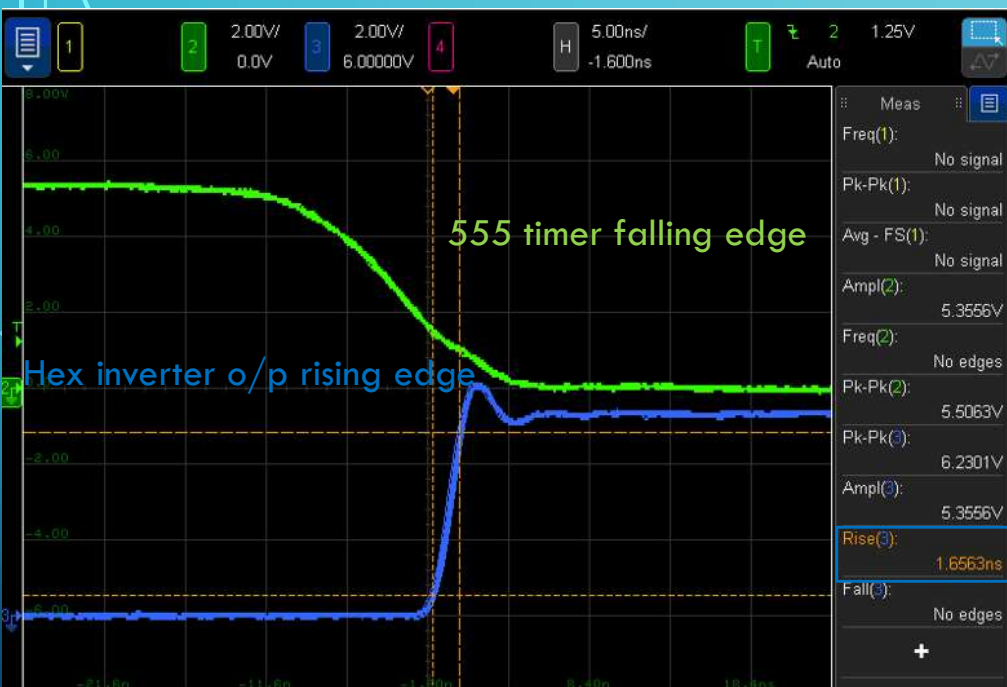
555 TIMER RISE & FALL TIME



Q HI NOISE



HEX INVERTER



CURRENT THROUGH LED



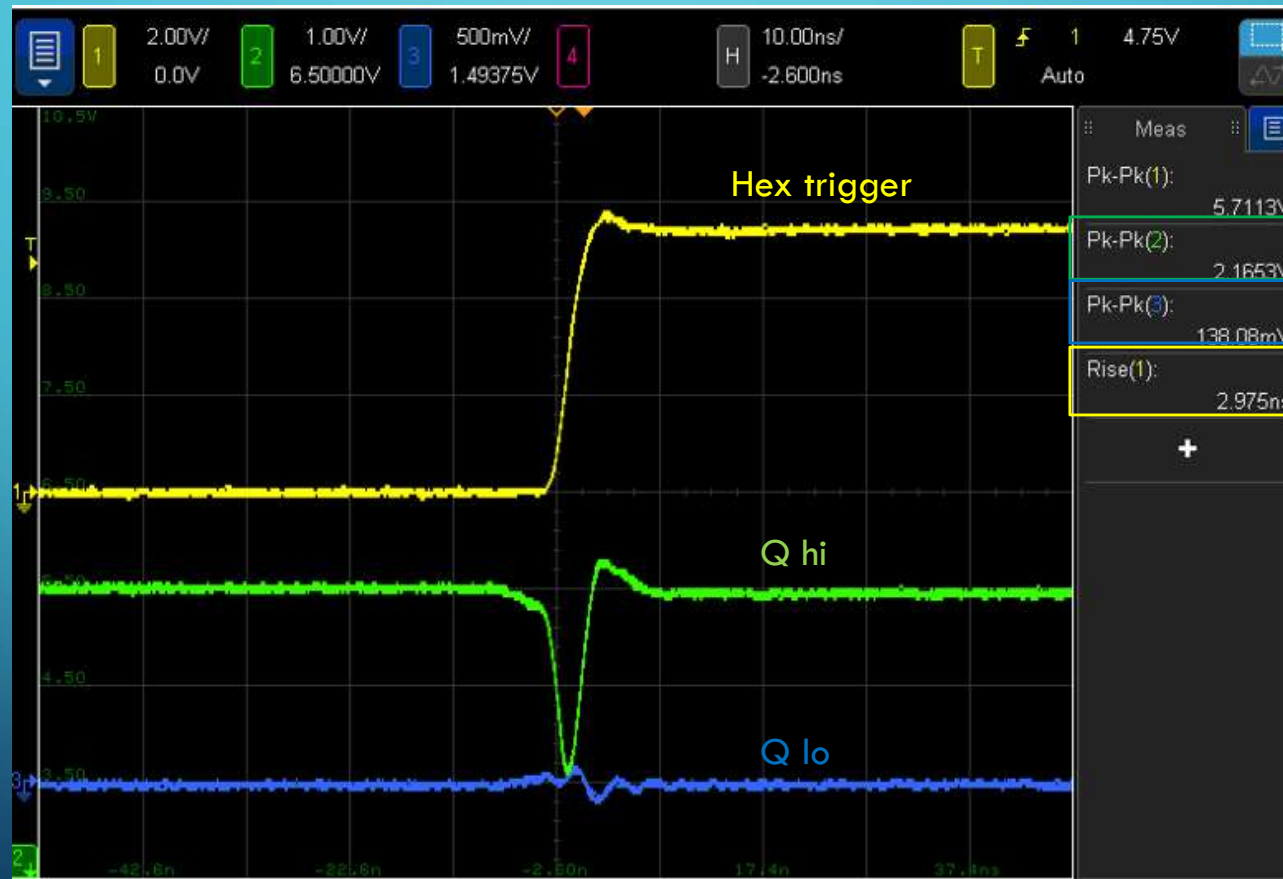
$$I = V/R$$
$$I = 1.81/47$$
$$I = 38.5\text{mA}$$

$$R_{th} = 47 * (5.3 - (2 + 1.81)) / (2 + 1.81)$$
$$R_{th} = 18.3\Omega$$

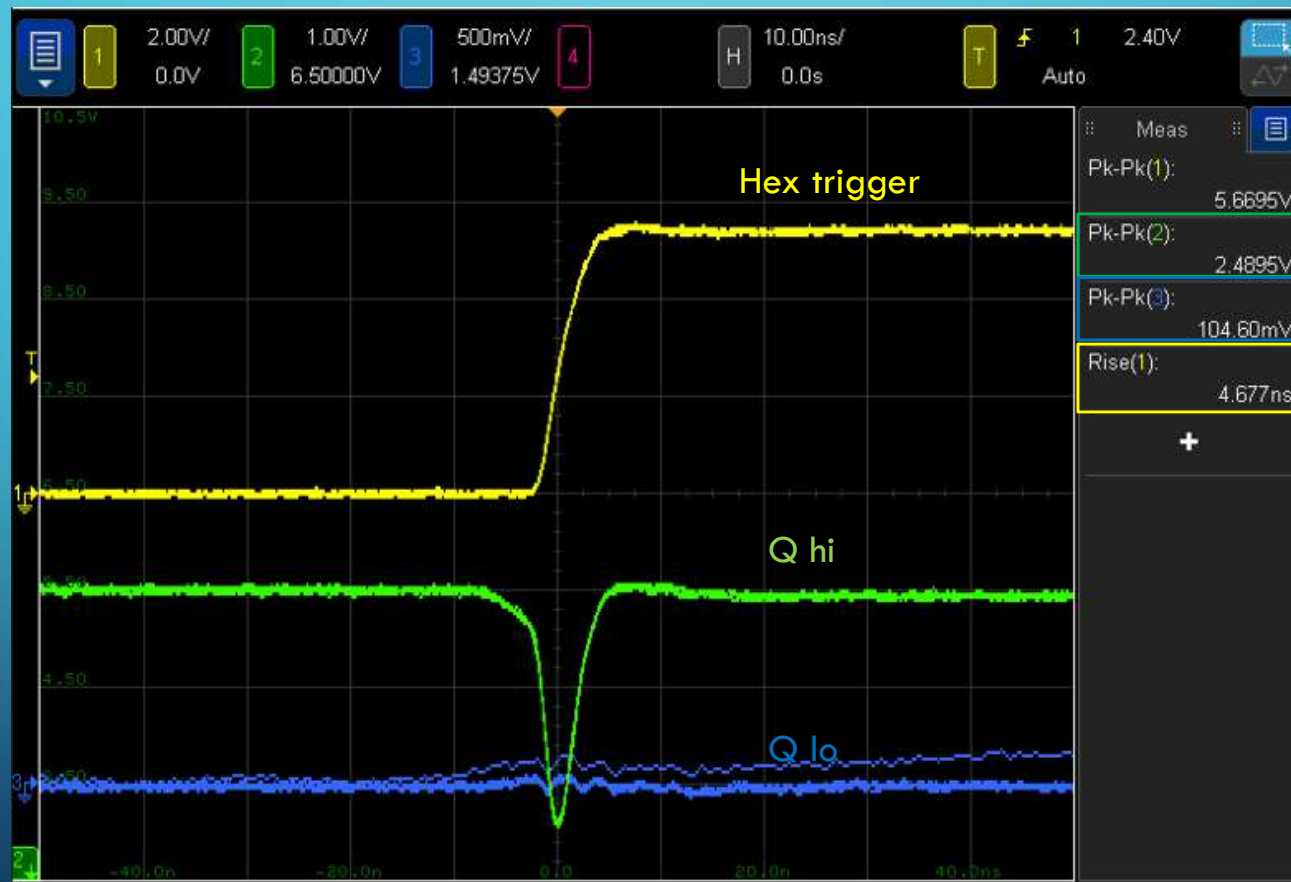
DECAP CLOSEBY:



2INCHES AWAY:



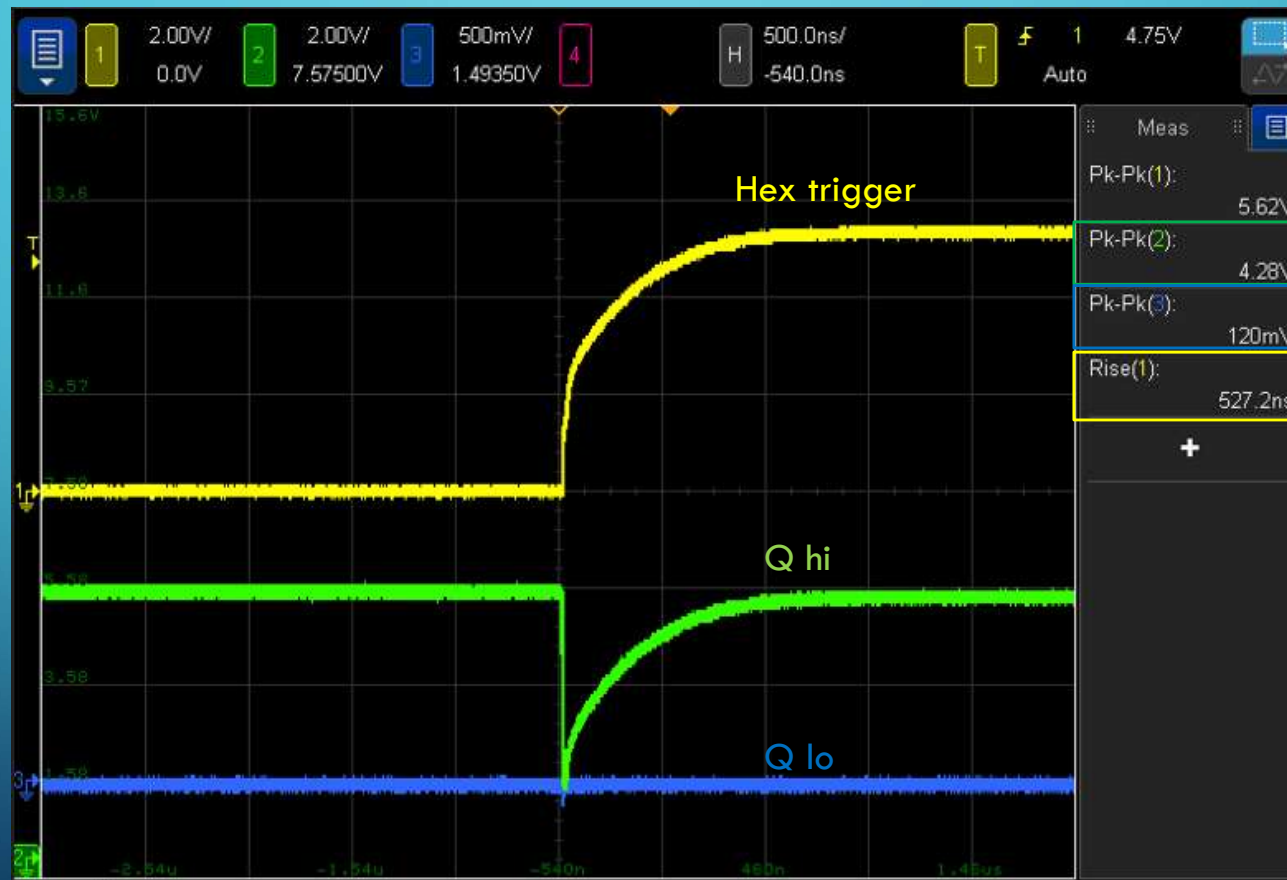
5INCHES AWAY:



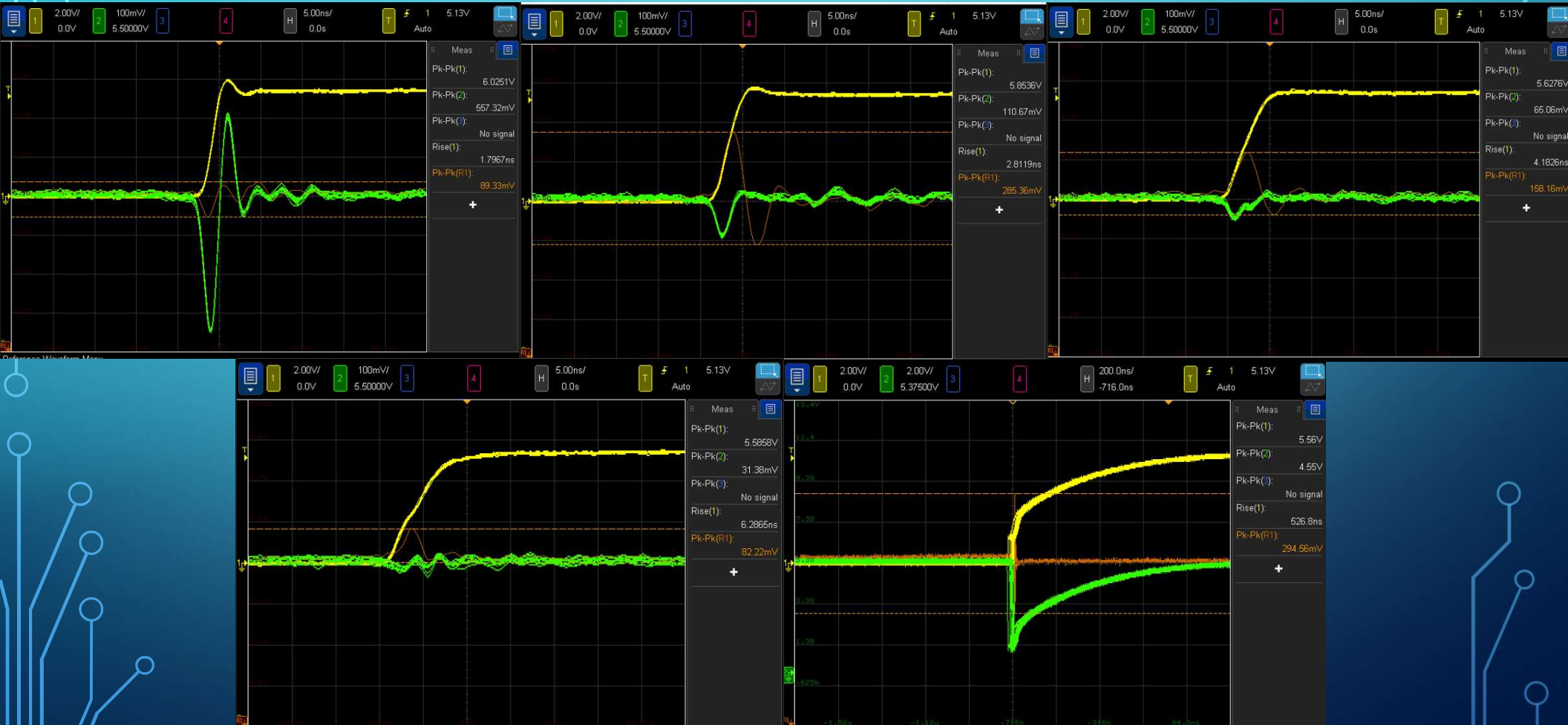
10 INCHES AWAY:



NO DECAP:



FERRITE BEAD FILTER EFFECT



COMPARISON

Decap placement	Rise time(nsec)	Q hi(V)	Q lo(mV)	Power rail Noise before ferrite(mV)	Power rail Noise after ferrite(mV)
Close	1.78	1.2	196	89	557
2 inch	2.98	2.16	138	285	110
5 inch	4.67	2.48	104	158	65
10 inch	7.9	2.87	138	82.22	31.38
No cap	527	4.28	120	294	4.55V

